

BEWARE

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Introduction

Folder overview

Code

```
## code/  
## |-- Step_pre_fisheries_statistics.R  
## |-- Step1_coding_treatments.R  
## |-- Step2_BEWARE_r1_0_0.R  
## |-- Step2b_iterBN.R  
## |-- Step3_scenarios.R  
## |-- Step3b_iter_scenarios.R  
## '-- supporting_r1.R
```

- Step_pre_fisheries_statistics.R: format large open source datasets into smaller .csv files #not run
- Step1_coding_treatments.R: format expert scores into a tidy file
- Step2_BEWARE_r1_0_0.R: handle all the conditional probability distributions and save a trained model

- Step2b_iterBN.R: iterates the script Step2_BEWARE_r1_0_0.R activating random samples and saving n trained models
- Step3_scenarios.R: test scenarios by setting evidences in decision nodes of the model produced by Step2_BEWARE_r1_0_0.R
- Step3b_iter_scenarios.R: test scenarios over the iterated n trained models produced by Step2b_iterBN.R
- supporting_r1.R: includes several custom functions

Data

```
## data/
## |-- editable_files/
## |   |-- networks/
## |   |   |-- BEWARE_release_v1_0_0.net
## |   |   |-- BEWARE_release_v1_0_0.xdsl
## |   |-- ~$dialogues_raw.xlsx
## |   |-- ~$questionnaires_coded.xlsx
## |   |-- adaptive_revised_v2.csv
## |   |-- coding_expert_revision_blind.xlsx
## |   |-- dialogues_raw.xlsx
## |   |-- nodes_text.xlsx
## |   |-- questionnaires_coded.xlsx
## |-- fisheries_statistics/
## |   |-- AER_SWE.csv
## |   |-- fdi_catch_SWE.csv
## |   |-- fdi_effort_SWE.csv
## |   |-- rec_catch_SWE.csv
## |   |-- rec_economics_SWE.csv
## |   |-- rec_gears_SWE.csv
## |-- read_only/
## |   |-- networks/
## |   |   |-- BEWARE_learnt_r1_0_0.net
## |   |-- coastal_catch.csv
## |   |-- coastal_fdays.csv
## |   |-- coding_report.csv
## |   |-- coding_report_unc.csv
## |   |-- cost_df.csv
## |   |-- cost_df_v2.csv
## |   |-- costs.csv
## |   |-- extra_nodes_cpt.csv
## |   |-- relevance.csv
```

- editable_files: contain all the files where the user can input values
 - networks: stores the BN templates
 - * BEWARE_release_v1_0_0.net: BN template in .net format, layout NOT supported by GeNIe
 - * BEWARE_release_v1_0_0.xdsl: BN template in .xdsl format, with layout supported by GeNIe
 - adaptive_revised_v2.csv: file indicating which answer refers to an adaptive strategy
 - coding_expert_revision_blind.xlsx # revision of unclear answers
 - dialogues_raw.xlsx

- nodes_text.xlsx
- questionnaires_coded.xlsx
- fisheries_statistics: files described in Supplementary Information B (docs folder)
- read_only: files automatically generated that does not need to be modified

Docs

```
## docs/
## |-- figure/
## |   |-- unnamed-chunk-10-1.pdf
## |   |-- unnamed-chunk-12-1.pdf
## |   |-- unnamed-chunk-14-1.pdf
## |   |-- unnamed-chunk-16-1.pdf
## |   |-- unnamed-chunk-18-1.pdf
## |   |-- unnamed-chunk-20-1.pdf
## |   |-- unnamed-chunk-27-1.pdf
## |   |-- unnamed-chunk-29-1.pdf
## |   |-- unnamed-chunk-31-1.pdf
## |   |-- unnamed-chunk-33-1.pdf
## |   |-- unnamed-chunk-35-1.pdf
## |   |-- unnamed-chunk-37-1.pdf
## |   |-- unnamed-chunk-39-1.pdf
## |   |-- unnamed-chunk-4-1.pdf
## |   |-- unnamed-chunk-41-1.pdf
## |   |-- unnamed-chunk-43-1.pdf
## |   |-- unnamed-chunk-45-1.pdf
## |   |-- unnamed-chunk-6-1.pdf
## |   '-- unnamed-chunk-8-1.pdf
## |-- images/
## |   '-- BN_outline.jpg
## |-- author-info-blocks.lua
## |-- references.bib
## |-- scholarly-metadata.lua
## |-- SuppInfo_A.log
## |-- SuppInfo_A.pdf
## |-- SuppInfo_A.Rmd
## |-- SuppInfo_A_child1_interpretation.Rmd
## |-- SuppInfo_A_child2_interviews.Rmd
## |-- SuppInfo_B.log
## |-- SuppInfo_B.pdf
## |-- SuppInfo_B.Rmd
## |-- SuppInfo_B_child1_gears.pdf
## |-- SuppInfo_B_child1_gears.Rmd
## |-- SuppInfo_B_child2_economics.log
## |-- SuppInfo_B_child2_economics.pdf
## |-- SuppInfo_B_child2_economics.Rmd
## |-- SuppInfo_C.pdf
## |-- SuppInfo_C.Rmd
## |-- SuppInfo_z.pdf
## '-- SuppInfo_z.Rmd
```

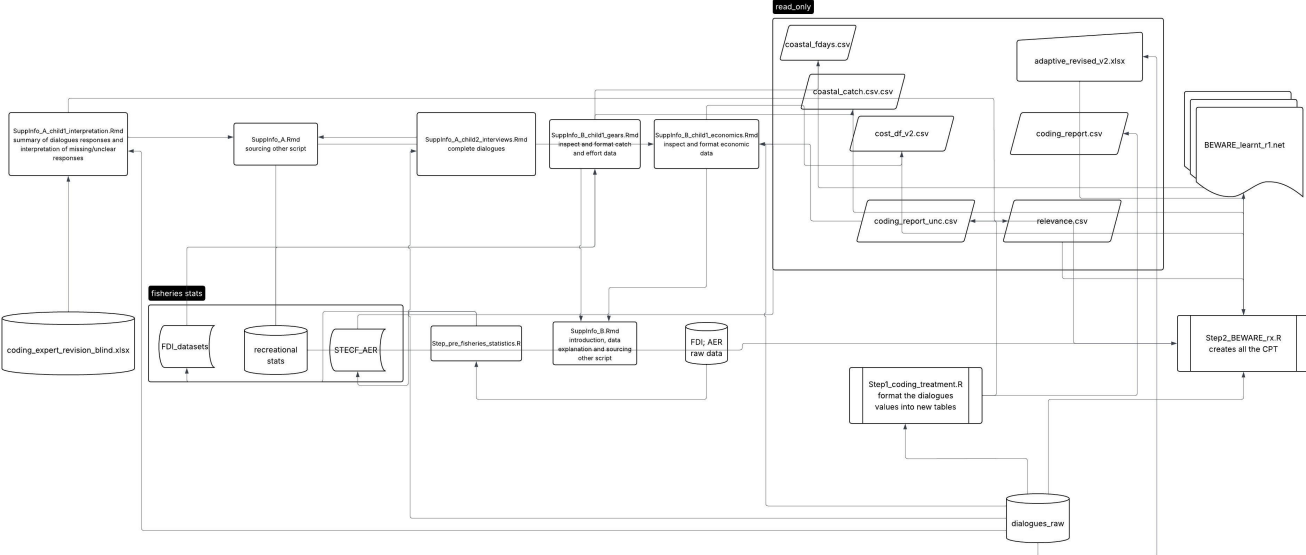


Figure 1: fvd

Include observations

The dataset `questionnaires_coded.xlsx` (Figure 2) is where all the dialogues answers are manually inserted. The dialogue text is reported in SuppInfo A (docs folder).

The fields that describes the answer metadata are:

- id:
- category
- id_q
- gear
- area
- target
- short_description

The other fields are used to report dialogues results. There are three type of questions:

- by event (e.g.: id_q 2-7): one value is expected for each event and it is reported in the columns a_hww; b_hws; c_ici; d_abl; e_erf; f_gal; g_sto according to the event
- one single value (e.g.: id_q16-18): the value is reported in the column *broad*
- one value with uncertainty (e.g.: id_q 11-12): the lower and upper range are included in the columns range.lwr and range.upr, the unit in unit_range

id	category	id_q	gear	area	target	broad	a_hww	b_hws	c_ici	d_abl	e_erf	f_gal	g_sto	range.lwr	range.upr	unit_range	short_description
1	demo	0				fishing guide											fishing_style
2	event	2	not_specified					4	1	1	1	5	1				
2	event	2	ice fishing				1										go_fishing
2	event	2	rod				5										
3	event	3	not_specified					2	2	2	-998	2	-998				
3	event	3	ice fishing				1										
3	event	3	rod				5										catchability
4	event	4					-997	-997	-998	-997	-998	2	-998				catch_assemblage
5	event	5					3	2	3	2	3	3	3				catch_condition
6	event	6					-999	3	-999	3	-999	-999	-999				cope_condition
7	event	7					4	-999	-999	-999	-999	-999	-999				lag_effect
8	event	8				1.5											unc_2_7
9	event	9	not_specified					0	2	0	1	1	2				personal_safety
9	event	9	ice fishing				1										
10	costs	10					0	0	2	0	2	1	2				damage
11	costs	11												10000	20000	sek_unit	minor_damage_cost
12	event	12												20000	6000000	sek_unit	major_damage_cost
13	event	13					-999	-999	-999	-999	-999	-999	-999				change_practices
14	event	14				1											unc_9_13
15	mobility	15					1	2	1	2	1	1	1				avoidance
16	mobility	16				0											gear characteristics
17	mobility	17				2											infrastructure
18	mobility	18				2											management_space

Figure 2: Figure 2: screenshot of one spreadsheet from the file questionnaires_coded.xlsx

Generate CPT

Scenarios